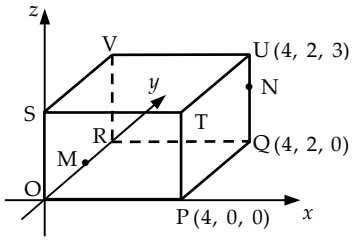
**1** The diagram shows a cuboid OPQR,STUV relative to the coordinate axes.
P is the point $(4, 0, 0)$,
Q is $(4, 2, 0)$ and U is $(4, 2, 3)$.
M is the midpoint of OR.

N is the point on UQ such that $UN = \frac{1}{3}UQ$.




(*a*) State the coordinates of M and N. **2**

(*b*) Express the vectors $\overrightarrow{VM}$ and $\overrightarrow{VN}$ in component form. **2**

**Treat as bad form, coordinates written as components and vice versa, throughout this question.**

| | Generic Scheme | | Illustrative Scheme | |
|---|---|---|---|---|
| **1 (a)** | | | | |
| $\bullet^1$ | ic | interpret midpoint for M | $\bullet^1$ | (0, 1, 0) |
| $\bullet^2$ | ic | interpret ratio for N | $\bullet^2$ | (4, 2, 2) |

| | | | | |
|---|---|---|---|---|
| **1 (b)** | | | | |
| $\bullet^3$ | ic | intepret diagram | $\bullet^3$ | $\overrightarrow{VM} = \begin{pmatrix} 0 \\ -1 \\ -3 \end{pmatrix}$ |
| $\bullet^4$ | pd | process vectors | $\bullet^4$ | $\overrightarrow{VN} = \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix}$ |

Using evidence from (a) or may have been taken directly from diagram.

**Notes**

1. V is the point $(0, 2, 3)$, which may or may not appear in the working to (b).

**Regularly occurring responses**

**Response 1**

(a) M(2, 0, 0) ✗ $\bullet^1$ N(4, 2, −1) ✗ $\bullet^2$

0 marks out of 2

(b) $\overrightarrow{VM} = \begin{pmatrix} 2 \\ -2 \\ -3 \end{pmatrix}$ ✗ $\bullet^3$

Consistent with (a)

$\overrightarrow{VN} = \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix}$ ✓ $\bullet^4$

From diagram

2 marks out of 2

**Response 2**

Incorrect V stated

(b) V(0, 3, 2)

$\overrightarrow{VM} = \begin{pmatrix} 0 \\ -2 \\ -2 \end{pmatrix}$ ✗ $\bullet^3$

$\overrightarrow{VN} = \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix}$ ✗ $\bullet^4$

1 mark out of 2

**Response 3**

(a) M(0, 2, 0) ✗ $\bullet^1$ N(4, 2, 2) ✓ $\bullet^2$

1 mark out of 2

(b) $\overrightarrow{VM} = \begin{pmatrix} -4 \\ 0 \\ -3 \end{pmatrix}$ ✗ $\bullet^3$

$\overrightarrow{VN} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}$ ✗ $\bullet^4$

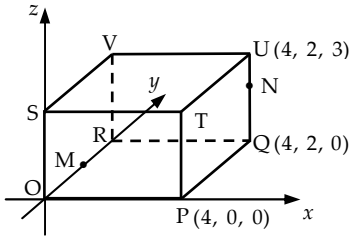
V(4, 2, 3) used in both but not stated

0 marks out of 2



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**1** The diagram shows a cuboid OPQR,STUV relative to the coordinate axes.
P is the point $(4, 0, 0)$,
Q is $(4, 2, 0)$ and U is $(4, 2, 3)$.
M is the midpoint of OR.
N is the point on UQ such that $UN = \frac{1}{3}UQ$.




(*c*) Calculate the size of angle MVN. **5**

**Treat as bad form, coordinates written as components and vice versa, throughout this question.**

**Generic Scheme** | **Illustrative Scheme**

**1 (c)**

**Method 1** : Vector Approach

| | | | |
|---|---|---|---|
| $\bullet^5$ | ss | know to use scalar product | $\cos M\hat{V}N = \dfrac{\overrightarrow{VM}\,.\,\overrightarrow{VN}}{\left\|\overrightarrow{VM}\right\|\left\|\overrightarrow{VN}\right\|}$ *stated, or implied by* $\bullet^9$ |
| $\bullet^6$ | pd | find scalar product | $\overrightarrow{VM}\,.\,\overrightarrow{VN} = 3$ |
| $\bullet^7$ | pd | find magnitude of a vector | $\left\|\overrightarrow{VM}\right\| = \sqrt{10}$ |
| $\bullet^8$ | pd | find magnitude of a vector | $\left\|\overrightarrow{VN}\right\| = \sqrt{17}$ |
| $\bullet^9$ | pd | evaluate angle | $76{\cdot}7^\circ$ or $1{\cdot}339$ rads or $85{\cdot}2$ grads |

**Method 2** : Cosine Rule Approach

| | | | |
|---|---|---|---|
| $\bullet^5$ | ss | know to use cosine rule | $\cos M\hat{V}N = \dfrac{VM^2 + VN^2 - MN^2}{2\times VM\times VN}$ *stated, or implied by* $\bullet^9$ |
| $\bullet^6$ | pd | find magnitude of a side | $VM = \sqrt{10}$ |
| $\bullet^7$ | pd | find magnitude of a side | $VN = \sqrt{17}$ |
| $\bullet^8$ | pd | find magnitude of a side | $MN = \sqrt{21}$ |
| $\bullet^9$ | pd | evaluate angle | $76{\cdot}7^\circ$ or $1{\cdot}339$ rads or $85{\cdot}2$ grads |

**Notes**

2. $\bullet^5$ is not available to candidates who choose to evaluate an incorrect angle.
3. For candidates who do not attempt $\bullet^9$, then $\bullet^5$ is only available if the formula quoted relates to the labelling in the question.
4. $\bullet^9$ should be awarded for any answer that rounds to $77^\circ$ or $1{\cdot}3$ rads or $85$ grads (i.e. correct to two significant figures.)

**Regularly occurring responses**

**Response 1**

$\cos M\hat{O}N = \dfrac{\overrightarrow{OM}\,.\,\overrightarrow{ON}}{\left\|\overrightarrow{OM}\right\|\left\|\overrightarrow{ON}\right\|}$ ✗ $\bullet^5$ — Wrong angle

$\overrightarrow{OM}\,.\,\overrightarrow{ON} = 2$ ✗ $\bullet^6$

$\left\|\overrightarrow{OM}\right\| = 1$ ✗ $\bullet^7$

$\left\|\overrightarrow{ON}\right\| = \sqrt{24}$ ✗ $\bullet^8$

Eased because only one non-zero component.

$65{\cdot}9^\circ$ or $1{\cdot}150$ rads or $73{\cdot}2$ grads ✗ $\bullet^9$

3 marks out of 5

**Response 2**

$\cos M\hat{V}N = \dfrac{\overrightarrow{VM}\,.\,\overrightarrow{VN}}{\left\|\overrightarrow{VM}\right\|\left\|\overrightarrow{VN}\right\|}$ ✓ $\bullet^5$

$\overrightarrow{VM}\,.\,\overrightarrow{VN} = 0$ ✗ $\bullet^6$

$\left\|\overrightarrow{VM}\right\| = \sqrt{17}$ ✗ $\bullet^7$

$\left\|\overrightarrow{VN}\right\| = 2$ ✗ $\bullet^8$

Going directly to $90^\circ$ from $\bullet^6$ would lose $\bullet^7$ and $\bullet^8$.

$90^\circ$ or equivalent ✗ $\bullet^9$

4 marks out of 5



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**2** (*a*) $12\cos x^\circ - 5\sin x^\circ$ can be expressed in the form $k\cos(x+a)^\circ$, where $k > 0$ and $0 \le a < 360$.
Calculate the values of *k* and *a*. **4**

**Generic Scheme** | **Illustrative Scheme**

**(a)**

| | | | | |
|---|---|---|---|---|
| $\bullet^1$ | ss | use addition formula | $\bullet^1$ $k\cos x^\circ \cos a^\circ - k\sin x^\circ \sin a^\circ$ or $k(\cos x^\circ \cos a^\circ - \sin x^\circ \cos a^\circ)$ | *stated explicitly* |
| $\bullet^2$ | ic | compare coefficients | $\bullet^2$ $k\cos a^\circ = 12$ and $k\sin a^\circ = 5$ or $-k\sin a^\circ = -5$ | *stated explicitly* |
| $\bullet^3$ | pd | process *k* | $\bullet^3$ 13 | *no justification required, but do not accept* $\sqrt{169}$ |
| $\bullet^4$ | pd | process *a* | $\bullet^4$ $22\cdot6$ | *accept any answer which rounds to* 23 |

**Notes**

1. Do not penalise the omission of the degree symbol.
2. Treat $k\cos x^\circ \cos a^\circ - \sin x^\circ \sin a^\circ$ as bad form only if the equations at the $\bullet^2$ stage both contain *k*.
3. $13\cos x^\circ \cos a^\circ - 13\sin x^\circ \sin a^\circ$ or $13(\cos x^\circ \cos a^\circ - \sin x^\circ \sin a^\circ)$ is acceptable for $\bullet^1$ and $\bullet^3$.
4. $\bullet^2$ is not available for $k\cos x^\circ = 12$ and $k\sin x^\circ = 5$ or $-k\sin x^\circ = -5$, however, $\bullet^4$ is still available.
5. $\bullet^4$ is lost to candidates who give *a* in radians only.
6. $\bullet^4$ may be gained only as a consequence of using evidence at $\bullet^2$ stage.
7. Candidates may use any form of the wave equation for $\bullet^1$, $\bullet^2$ and $\bullet^3$, however $\bullet^4$ is only available if the value of *a* is interpreted for the form $k\cos(x+a)^\circ$.

**Regularly occurring responses**

**Response 1A**

$k(\cos x^\circ \cos a^\circ - \sin x^\circ \sin a^\circ)$ ✓ $\bullet^1$

$\sin a = 5$ ✗ $\bullet^2$

$\cos a = 12$

$\tan a^\circ = \frac{5}{12}$

$a = 22\cdot6$ ✓✗ $\bullet^4$

$13\cos(x + 22\cdot6)$

✓ $\bullet^3$

2 marks out of 4

**Response 1B**

$k\cos x\cos a - k\sin x\sin a$ ✓ $\bullet^1$

$k = 13$ ✓ $\bullet^3$ ∧ $\bullet^2$

$\tan a^\circ = \frac{5}{12}$

$a = 22\cdot6$ ✓✗ $\bullet^4$

2 marks out of 4

**Response 2**

$k\cos(x-a)$

$= k\cos x\cos a + k\sin x\sin a$ ✓✗ $\bullet^1$

$= 13\cos x\cos a + 13\sin x\sin a$ ✓ $\bullet^3$

$13\cos a = 12$ $13\sin a = -5$ ✓✗ $\bullet^2$

then $a = 22\cdot6$ ✗ $\bullet^4$ — See note 6

or $a = 337\cdot4$ ✗ $\bullet^4$ — See note 7

3 marks out of 4

**Response 3A**

$k\cos x^\circ \cos a^\circ - \sin x^\circ \sin a^\circ$

$k\sin a = 5$ ✓ $\bullet^1$ ✓ $\bullet^2$

$k\cos a = 12$

$k = 13$ ✓ $\bullet^3$ $\tan a^\circ = \frac{12}{5}$ ✗ $\bullet^4$

$a = 67\cdot4$

3 marks out of 4

**Response 3B**

$k\cos x^\circ \cos a^\circ - \sin x^\circ \sin a^\circ$

$k\sin a = 12$ ✓ $\bullet^1$ ✗ $\bullet^2$

$k\cos a = 5$

$k = 13$ ✓ $\bullet^3$ $\tan a^\circ = \frac{12}{5}$

$a = 67\cdot4$ ✓✗ $\bullet^4$

3 marks out of 4

**Response 4**

$k\cos x^\circ \cos a^\circ - k\sin x^\circ \sin a^\circ$ ✓ $\bullet^1$

$k\cos a = 12$ ✗ $\bullet^2$

$k\sin a = -5$

$k = 13$ ✓ $\bullet^3$ $\tan a^\circ = -\frac{5}{12}$ — See note 6

$a = 337\cdot4$ ✓✗ $\bullet^4$

3 marks out of 4



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**2** (*b*) (i) Hence state the maximum and minimum values of $12\cos x^\circ - 5\sin x^\circ$.
(ii) Determine the values of $x$, in the interval $0 \le x < 360$, at which these maximum and minimum values occur. **3**

| Generic Scheme | | | Illustrative Scheme | |
|---|---|---|---|---|
| **(b)** | | | | |
| $\bullet^5$ | ss | state maximum and minimum | $\bullet^5$ | $13,\ -13$ |
| $\bullet^6$ | ic | find $x$ corresponding to max. value | $\bullet^6$ | maximum at $337\cdot4$ and no others |
| $\bullet^7$ | pd | find $x$ corresponding to min. value | $\bullet^7$ | minimum at $157\cdot4$ and no others |
| | | | or | |
| | | | $\bullet^6$ | $337\cdot4$ and $157\cdot4$ and no others |
| | | | $\bullet^7$ | maximum at $337\cdot4$ and minimum at $157\cdot4$ |

**Notes**

8. $\bullet^5$ is available for $\sqrt{169}$ and $-\sqrt{169}$ only if $\sqrt{169}$ has been penalised at $\bullet^3$.
9. Accept answers which round to 337 and 157 for $\bullet^6$ and $\bullet^7$.
10. Candidates who continue to work in radian measure should not be penalised further.
11. Extra solutions, correct or incorrect, should be penalised at $\bullet^6$ or $\bullet^7$ but not both.
12. $\bullet^6$ and $\bullet^7$ are not available to candidates who work with $13\cos(x+22\cdot6)^\circ = 0$ or $13\cos(x+22\cdot6)^\circ = 1$.
13. Candidates who use $13\cos(x-22\cdot6)^\circ$ from a correct (a) lose $\bullet^6$ but $\bullet^7$ is still available.

**Regularly occurring responses**

**Response 1**

From (a) $a = 67\cdot4$

max/ min $= \pm13$ ✓ $\bullet^5$

max at $292\cdot6$ ✗ $\bullet^6$

min at $112\cdot6$ ✗ $\bullet^7$

3 marks out of 3

**Response 2**

From (a) $\sqrt{169}\cos(x+22\cdot6)^\circ$

max $= \sqrt{169}$ min $= -\sqrt{169}$ ✗ $\bullet^5$

max at $22\cdot6$ ✗ $\bullet^6$

min at $202\cdot6$ ✗ $\bullet^7$

2 marks out of 3

$\sqrt{169}$ already penalised at (a)

**Response 3A**

$13\cos(x+22\cdot6)^\circ$ (×13, $22\cdot6$ →)

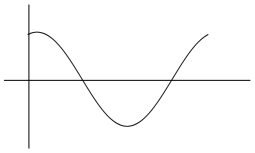
max at $22\cdot6$ ✗ $\bullet^6$

min at $202\cdot6$ ✗ $\bullet^7$

1 mark out of 3

Insufficient evidence for $\bullet^5$

**Response 3B**



$22\cdot6$ →

$13\cos(x+22\cdot6)^\circ$ max at $22\cdot6$ ✗ $\bullet^6$

min at $202\cdot6$ ✗ $\bullet^7$

1 mark out of 3

**N.B.** Candidates who use differentiation in (b) can gain $\bullet^5$ only, as a direct result of their response in (a).
This question is in degrees and so calculus is not appropriate for $\bullet^6$ and $\bullet^7$.



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**3** (*a*) (i) Show that the line with equation $y = 3 - x$ is a tangent to the circle with equation $x^2 + y^2 + 14x + 4y - 19 = 0$.

(ii) Find the coordinates of the point of contact, P. **5**

| Generic Scheme | | | Illustrative Scheme |
|---|---|---|---|
| **(a)** | | | |
| $\bullet^1$ | ss | substitute | $\bullet^1$ $x^2 + (3-x)^2 + 14x + 4(3-x) - 19 = 0$ |
| | | | **Method 1** : Factorising |
| $\bullet^2$ | pd | express in standard form | $\bullet^2$ $2x^2 + 4x + 2$ } |
| $\bullet^3$ | ic | start proof | $\bullet^3$ $2(x+1)(x+1)$ } $= 0$ *see note* 1 |
| $\bullet^4$ | ic | complete proof | $\bullet^4$ equal roots so line is a tangent |
| | | | **Method 2** : Discriminant |
| | | | $\bullet^2$ $2x^2 + 4x + 2 = 0$ *stated explicitly* |
| | | | $\bullet^3$ $4^2 - 4 \times 2 \times 2$ |
| | | | $\bullet^4$ $b^2 - 4ac = 0$ so line is a tangent |
| $\bullet^5$ | pd | coordinates of P | $\bullet^5$ $x = -1,\ y = 4$ |

**Notes**

**For method 1 :**

1. $\bullet^2$ is only available if "= 0" appears at either $\bullet^2$ or $\bullet^3$ stage.
2. Alternative wording for $\bullet^4$ could be e.g. 'repeated roots', 'repeated factor', ' only one solution', 'only one point of contact' **along with** 'line is a tangent'.

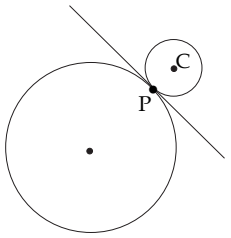
**For both methods :**

3. Candidates must work with a quadratic equation at the $\bullet^3$ and $\bullet^4$ stages.
4. Simply stating the tangency condition without supporting working cannot gain $\bullet^4$.
5. For candidates who obtain two distinct roots, $\bullet^4$ is still available for 'not equal roots so not a tangent' or '$b^2 - 4ac \neq 0$ so line is not a tangent', but $\bullet^5$ is not available.



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**3** (*b*) Relative to a suitable set of coordinate axes, the diagram below shows the circle from (*a*) and a second smaller circle with centre C.




The line $y = 3 - x$ is a common tangent at the point P.

The radius of the larger circle is three times the radius of the smaller circle.

Find the equation of the smaller circle. **6**

$x^2 + y^2 + 14x + 4y - 19 = 0$

**Generic Scheme** **Illustrative Scheme**

**(b)**

**Method 1** : via centre and radius

- $\bullet^6$ ic state centre of larger circle
- $\bullet^7$ ss find radius of larger circle
- $\bullet^8$ pd find radius of smaller circle
- $\bullet^9$ ss strategy for finding centre
- $\bullet^{10}$ ic interpret centre of smaller circle
- $\bullet^{11}$ ic state equation

**Method 2** : via ratios

- $\bullet^6$ ic state centre of larger circle
- $\bullet^7$ ss strategy for finding centre
- $\bullet^8$ ic state centre of smaller circle
- $\bullet^9$ ss strategy for finding radius
- $\bullet^{10}$ pd find radius of smaller circle
- $\bullet^{11}$ ic state equation

**Method 1** : via centre and radius

- $\bullet^6$ $(-7, -2)$ *see note* 11
- $\bullet^7$ $\sqrt{72}$ *see note* 6 *stated, or implied by* $\bullet^8$
- $\bullet^8$ $\sqrt{8}$ *see note* 7
- $\bullet^9$ e.g "Stepping out"
- $\bullet^{10}$ (1, 6)
- $\bullet^{11}$ $(x-1)^2 + (y-6)^2 = 8$ or $x^2 + y^2 - 2x - 12y + 29 = 0$

**Method 2** : via ratios

- $\bullet^6$ $(-7, -2)$ *see note* 11
- $\bullet^7$ e.g. "Stepping out"
- $\bullet^8$ (1, 6)
- $\bullet^9$ $\sqrt{2^2 + 2^2}$
- $\bullet^{10}$ $\sqrt{8}$ *see note* 10
- $\bullet^{11}$ $(x-1)^2 + (y-6)^2 = 8$ or $x^2 + y^2 - 2x - 12y + 29 = 0$

**Notes**

**For method 1**:

6. Acceptable alternatives for $\bullet^7$ are $6\sqrt{2}$ or decimal equivalent which rounds to $8 \cdot 5$ i.e. to two significant figures.
7. Acceptable alternatives for $\bullet^8$ are $\frac{\sqrt{72}}{3}$ or $2\sqrt{2}$ or decimal equivalent which rounds to $2 \cdot 8$.
8. (1, 6) without working gains $\bullet^{10}$ but loses $\bullet^9$.

**For method 2**:

9. (1, 6) without working gains $\bullet^8$ but loses $\bullet^7$.
10. Acceptable alternatives for $\bullet^{10}$ are $2\sqrt{2}$ or decimal equivalent which rounds to $2 \cdot 8$.

**In both methods**:

11. If $m = 1$ is used in a 'stepping out' method the centre of the larger circle need not be stated explicitly for $\bullet^6$.
12. For the smaller circle, candidates who 'guess' values for either the centre or radius cannot be awarded $\bullet^{11}$.
13. At $\bullet^{11}$ e.g. $\sqrt{8}^2$, $2 \cdot 8^2$ are unacceptable, but any decimal which rounds to $7 \cdot 8$ is acceptable.
14. $\bullet^{11}$ is not available to candidates who divide the coordinates of the centre of the larger circle by 3.



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**4** Solve $2\cos 2x - 5\cos x - 4 = 0$ for $0 \leq x < 2\pi$. **5**

**Generic Scheme** **Illustrative Scheme**

**4**

$\bullet^1$ ss know to use double angle formula
$\bullet^2$ ic express as quadratic in $\cos x$
$\bullet^3$ ss start to solve

**Method 1** : Using factorisation

$\bullet^1$ $2\times(2\cos^2 x - 1)\ldots$
$\bullet^2$ $4\cos^2 x - 5\cos x - 6$ } = 0 must appear at either of
$\bullet^3$ $(4\cos x + 3)(\cos x - 2)$ } these lines to gain $\bullet^2$.

**Method 2** : Using quadratic formula

$\bullet^1$ $2\times(2\cos^2 x - 1)\ldots$
$\bullet^2$ $4\cos^2 x - 5\cos x - 6 = 0$
$\bullet^3$ $\cos x = \dfrac{-(-5) \pm \sqrt{(-5)^2 - 4\times 4\times(-6)}}{2\times 4}$

**In both methods** :

$\bullet^4$ pd reduce to equations in $\cos x$ only
$\bullet^5$ pd complete solutions to include only one where $\cos x = k$ with $|k| > 1$

$\bullet^4$ $\cos x = -\frac{3}{4}$ and $\cos x = 2$
$\bullet^5$ $2\cdot 419,\ 3\cdot 864$ and no solution

**or**

$\bullet^4$ $\cos x = 2$ and no solution
$\bullet^5$ $\cos x = -\frac{3}{4}$ and $2\cdot 419,\ 3\cdot 864$

**Notes**

1. $\bullet^1$ is not available for simply stating that $\cos 2A = 2\cos^2 A - 1$ with no further working.
2. Substituting $\cos 2A = 2\cos^2 A - 1$ or $\cos 2a = 2\cos^2 a - 1$ etc. should be treated as bad form throughout.
3. In the event of $\cos^2 x - \sin^2 x$ or $1 - 2\sin^2 x$ being substituted for $\cos 2x$, $\bullet^1$ cannot be given until the equation reduces to a quadratic in $\cos x$.
4. Candidates may express the quadratic equation obtained at the $\bullet^2$ stage in the form $4c^2 - 5c + 6 = 0$, $4x^2 - 5x + 6 = 0$ etc. For candidates who do not solve a trig. equation at $\bullet^5$, $\cos x$ must appear explicitly to gain $\bullet^4$.
5. $\bullet^4$ and $\bullet^5$ are only available as a consequence of solving a quadratic equation subsequent to a substitution.
6. Any attempt to solve $a\cos^2 x + b\cos x = c$ loses $\bullet^3$, $\bullet^4$ and $\bullet^5$.
7. Accept answers given as decimals which round to $2\cdot 4$ and $3\cdot 9$.
8. There must be an indication after $\cos x = 2$ that there are no solutions to this equation.

   Acceptable evidence : e.g. "~~$\cos x = 2$~~" , "NA", "out of range", "invalid" and "$\cos x = 2$ no", "$\cos x = 2$ ✗"

   Unacceptable evidence : e.g. "$\underline{\underline{\cos x = 2}}$", "$\cos x = 2$ ???", "Maths Error".

9. $\bullet^5$ is not available to candidates who work throughout in degrees and do not convert their answer into radian measure.
10. Do not accept e.g. $221\cdot 4,\ 138\cdot 6,\ \dfrac{221\cdot 4\pi}{180},\ \dfrac{221\pi}{180},\ 1\cdot 23\pi$.
11. Ignore correct solution outside the interval $0 \leq x < 2\pi$.



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**4**

**Regularly occurring responses**

**Response 1**

$2\times 2\cos^2 x - 1 \ldots$ ✓ $\bullet^1$

$4\cos^2 x - 5\cos x - 5 = 0$ ✗ $\bullet^2$

$\cos x = \dfrac{-(-5) \pm \sqrt{(-5)^2 - 4\times 4\times (-5)}}{2\times 4}$ ✓ $\bullet^3$

$\cos x = \dfrac{5-\sqrt{105}}{8}$ and $\cos x = \dfrac{5+\sqrt{105}}{8}$ ✓ $\bullet^4$

$2\cdot 286,\ 3\cdot 997$ and no solution ✓ $\bullet^5$

$\bullet^5$ is only available to candidates where one, but not both, of their equations has no solution for $\cos x$.

4 marks out of 5

---

**Response 2**

$4\cos^2 x - 1 \ldots$ ✓ $\bullet^1$

$4\cos^2 x - 5\cos x - 5 = 0$ ✗ $\bullet^2$

$(2\cos x + 1)(2\cos x - 5) = 0$ ✗ $\bullet^3$

$\cos x = -\dfrac{1}{2}$ and ~~$\cos x = \dfrac{5}{2}$~~ ✓ $\bullet^4$

$x = \dfrac{2\pi}{3},\ \dfrac{4\pi}{3}$ ✓ $\bullet^5$

$4\cos^2 x - 1$ with no further working cannot gain $\bullet^1$; however if a quadratic in $\cos x$ subsequently appears then $\bullet^1$ is awarded but $\bullet^2$ is not available.

3 marks out of 5

---

$\bullet^1$ is lost here as it is not clear whether the candidate has used $2\cos^2 x - 1$ or $\cos^2 x - 1$ as their substitution.

**Response 3A**

$2\cos^2 x - 1 - 5\cos x - 4 = 0$ ✗ $\bullet^1$

$2\cos^2 x - 5\cos x - 5 = 0$ ✓ $\bullet^2$

$\cos x = \dfrac{5 \pm \sqrt{25+40}}{4}$ ✓ $\bullet^3$

$\cos x = -0\cdot 766$ and $\cos x = 3\cdot 267$ ✓ $\bullet^4$

$x = 2\cdot 44,\ 3\cdot 84$ and undefined ✓ $\bullet^5$

4 marks out of 5

**Response 3B**

$\cos 2x = 2\cos^2 x - 1$

$2\cos^2 x - 1 - 5\cos x - 4 = 0$ ✓ $\bullet^1$

$2\cos^2 x - 5\cos x - 5 = 0$ ✗ $\bullet^2$

$\cos x = \dfrac{5 \pm \sqrt{25+40}}{4}$ ✓ $\bullet^3$

$\cos x = -0\cdot 766$ and $\cos x = 3\cdot 267$ ✓ $\bullet^4$

$x = 2\cdot 44,\ 3\cdot 84$ and undefined ✓ $\bullet^5$

4 marks out of 5

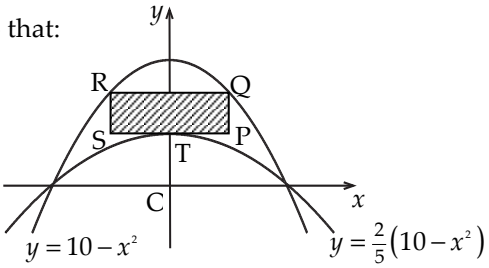


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**5** The parabolas with equations $y = 10 - x^2$ and $y = \frac{2}{5}(10 - x^2)$ are shown in the diagram below.

A rectangle PQRS is placed between the two parabolas as shown, so that:

- Q and R lie on the upper parabola.
- RQ and SP are parallel to the $x$-axis.
- T , the turning point of the lower parabola, lies on SP.




($a$) (i) If TP = $x$ units, find an expression for the length of PQ.

(ii) Hence show that the area, $A$, of rectangle PQRS is given by

$$A(x) = 12x - 2x^3$$ **3**

($b$) Find the maximum area of this rectangle. **6**

**Generic Scheme** **Illustrative Scheme**

**5 (a)**

| | | | | |
|---|---|---|---|---|
| $\bullet^1$ | ss | know to and find OT | $\bullet^1$ | 4 or (0, 4) *stated, or implied by* $\bullet^2$ |
| $\bullet^2$ | ic | obtain an expression for PQ | $\bullet^2$ | $10 - x^2 - 4$ |
| $\bullet^3$ | ic | complete area evaluation | $\bullet^3$ | $2x \times (6 - x^2) = 12x - 2x^3$ |

**Notes**

1. The evidence for $\bullet^1$ and $\bullet^2$ may appear on a sketch.
2. No marks are available to candidates who work backwards from the area formula.
3. $\bullet^3$ is only available if $\bullet^2$ has been awarded.

**5 (b)**

| | | | | |
|---|---|---|---|---|
| $\bullet^4$ | ss | know to and start to differentiate | $\bullet^4$ | $A'(x) = 12 \ldots$ *stated, or implied by* $\bullet^5$ |
| $\bullet^5$ | pd | complete differentiation | $\bullet^5$ | $12 - 6x^2$ |
| $\bullet^6$ | ic | set derivative to zero | $\bullet^6$ | $12 - 6x^2 = 0$ |
| $\bullet^7$ | pd | obtain $x$ | $\bullet^7$ | $\sqrt{2}$ or decimal equivalent $\left(\text{ignore inclusion of } -\sqrt{2}\right)$ |
| $\bullet^8$ | ss | justify nature of stationary point | $\bullet^8$ | $x$ \| … \| $\sqrt{2}$ \| …<br>$A'(x)$ \| + \| 0 \| −<br>(Note : accept $12 - 6x^2$ in lieu of $A'(x)$ in the nature table.) |
| $\bullet^9$ | ic | interpret result and evaluate area | $\bullet^9$ | Max **and** $8\sqrt{2}$ or decimal equivalent<br>N.B. To conclude a maximum the evidence must come from $\bullet^8$. |

**Notes**

4. At $\bullet^7$ accept any answer which rounds to $1 \cdot 4$.
5. Throughout this question treat the use of $f'(x)$ or $\frac{dy}{dx}$ as bad form.
6. At $\bullet^8$ the nature can be determined using the second derivative.
7. At $\bullet^9$ accept any answer which rounds to $11 \cdot 3$ or $11 \cdot 4$.



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**5**

**Regularly occurring responses**

**Response 1**

$A(x) = 12x - 2x^3$

$A'(x) = 24x^2 - 6x^3$ ✗ •4 ✗ •5

$24x^2 - 6x^3 = 0$ ✗ •6

$A'(x) = 0$ on its own would not be sufficient for •6.

**Response 2**

At stationary points, $A'(x) = 0$

$12 - 6x^2$ ✓ •4 ✓ •5 ✓ •6

$x = \sqrt{2}$ ✓ •7

**Response 3**

$A(x) = 12x - 2x^3$

$= 12 - 6x^2$ ✓ •4 ✓ •5

Bad form

**Response 4A**

| | ... | $\sqrt{2}$ | ... |
|---|---|---|---|
| $A'(x)$ | + | 0 | − |

✗ •8

$x$ missing

**Response 4B**

| $x$ | 1 | $\sqrt{2}$ | 2 |
|---|---|---|---|
| $A'(x)$ | 6 | 0 | −12 |

✓ •8

**Response 4C**

| $x$ | ... | $\sqrt{2}$ | ... |
|---|---|---|---|
| $A'(x)$ | \ | — | / |

✗ •8

or 'slope' etc.

signs or values are necessary

**Response 5**

Maximum at $x = \sqrt{2}$

$y = 12\sqrt{2} - 2\sqrt{2}^3 = 8\sqrt{2}$ ✓ •9

Area $= 2\sqrt{2} \times 8\sqrt{2} = 32$

Treat this as an error subsequent to a correct answer.



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**6** (*a*) A curve has equation $y = (2x-9)^{\frac{1}{2}}$.

Show that the equation of the tangent to this curve at the point where $x = 9$ is $y = \frac{1}{3}x$. **5**

(*b*) Diagram 1 shows part of the curve and the tangent.

The curve cuts the $x$-axis at the point A.

Find the coordinates of point A. **1**

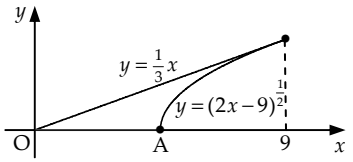


*Diagram 1*

**Generic Scheme** **Illustrative Scheme**

**6 (a)**

| | | | |
|---|---|---|---|
| $\bullet^1$ | ss | know to and start to differentiate | $\bullet^1$ $\frac{1}{2}(2x-9)^{-\frac{1}{2}}$ |
| $\bullet^2$ | pd | complete chain rule derivative | $\bullet^2$ $\ldots\times 2$ |
| $\bullet^3$ | pd | gradient via differentiation | $\bullet^3$ $\frac{1}{3}$ |
| $\bullet^4$ | pd | obtain $y_{\text{CURVE}}$ at $x = 9$ | $\bullet^4$ 3 |
| $\bullet^5$ | ic | state equation and complete | $\bullet^5$ $y-3=\frac{1}{3}(x-9)$ and complete to $y=\frac{1}{3}x$ |

**Notes**

1. $\bullet^3$ is only available as a consequence of differentiating equation of the curve.
2. Candidates must arrive at the equation of the tangent via the point (9, 3) and not the origin.
3. For $\bullet^3$ accept $9^{-\frac{1}{2}}$.

**Regularly occurring responses**

**Response 1**
Candidates who equate derivatives:

$\frac{1}{2}(2x-9)^{-\frac{1}{2}}\times 2=\frac{1}{3}$ ✓ $\bullet^3$
✓ $\bullet^1$ ✓ $\bullet^2$

leading to $x = 9$ and $y = 3$ from curve ✓ $\bullet^4$
Also obtaining $y = 3$ from line and so line is a tangent ✓ $\bullet^5$

5 marks out of 5

**Response 2**
Candidates who intersect curve and line:

$(2x-9)^{\frac{1}{2}}=\frac{1}{3}x$ ✓ $\bullet^1$

$2x-9=\left(\frac{1}{3}x\right)^2$ ✓ $\bullet^2$

$\frac{1}{9}x^2-2x+9=0$ ✓ $\bullet^3$

5 marks out of 5

Factorising or using discriminant ✓ $\bullet^4$
Equal roots or $b^2-4ac=0$ so line is a tangent ✓ $\bullet^5$

**(b)**

| | | | |
|---|---|---|---|
| $\bullet^6$ | ic | obtain coordinates of A | $\bullet^6$ $\left(\frac{9}{2}, 0\right)$ |

**Notes**

4. Accept $x=\frac{9}{2}$, $y=0$ where $y=0$ may appear from $(2x-9)^{\frac{1}{2}}=0$.
5. For $\left(\frac{9}{2}, 0\right)$ without working $\bullet^6$ is awarded, but from erroneous working $\bullet^6$ is lost (see response below).

**Regularly occurring responses**

**Response 1**

$\sqrt{2x-9}=0 \Rightarrow \sqrt{2x}-3=0 \Rightarrow 2x-9=0 \Rightarrow x=4{\cdot}5$

Here $\bullet^6$ cannot be awarded due to the erroneous working.



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**6** *(c)* Calculate the shaded area shown in diagram 2. **7**

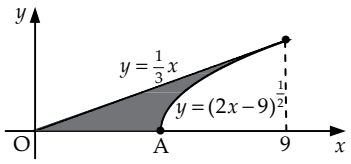



*Diagram 2*

**Generic Scheme** **Illustrative Scheme**

**6 (c)**

**Method 1** : Area of triangle – area under curve

| | | | |
|---|---|---|---|
| $\bullet^7$ | ss | strategy for finding shaded area | Shaded area = Area of large $\Delta$ – Area under curve |
| $\bullet^8$ | ss | know to integrate $(2x-9)^{\frac{1}{2}}$ | $\int (2x-9)^{\frac{1}{2}}\,dx$ |
| $\bullet^9$ | pd | start integration | $\frac{(2x-9)^{\frac{3}{2}}}{\frac{3}{2}}$ |
| $\bullet^{10}$ | pd | complete integration | $\ldots\times\frac{1}{2}$ |
| $\bullet^{11}$ | ic | limits $x_A$ and 9 | $\frac{9}{2}$ and 9 |
| $\bullet^{12}$ | pd | substitute limits | $\frac{1}{3}(18-9)^{\frac{3}{2}}-0$ |
| $\bullet^{13}$ | pd | evaluate area and complete strategy | $\frac{27}{2}-9=\frac{9}{2}$ or $4\frac{1}{2}$ or $4\cdot5$ |

**Method 2** : Area between line and curve

| | | | |
|---|---|---|---|
| $\bullet^7$ | ss | strategy for finding shaded area | Area of small $\Delta$ + area between line and curve |
| $\bullet^8$ | ss | know to integrate $(2x-9)^{\frac{1}{2}}$ | $\int \ldots (2x-9)^{\frac{1}{2}}\,dx$ |
| $\bullet^9$ | pd | start integration | $\ldots\frac{(2x-9)^{\frac{3}{2}}}{\frac{3}{2}}$ |
| $\bullet^{10}$ | pd | complete integration | $\ldots\times\frac{1}{2}$ |
| $\bullet^{11}$ | ic | limits $x_A$ and 9 | $\frac{9}{2}$ and 9 |
| $\bullet^{12}$ | pd | 'upper − lower' and substitute limits | $\left(\frac{1}{6}\times9^2-\frac{1}{3}(18-9)^{\frac{3}{2}}\right)-\left(\frac{1}{6}\times\left(\frac{9}{2}\right)^2-\frac{1}{3}(9-9)^{\frac{3}{2}}\right)$ |
| $\bullet^{13}$ | pd | evaluate area and complete strategy | $\frac{27}{8}+\frac{9}{8}=\frac{9}{2}$ or $4\frac{1}{2}$ or $4\cdot5$ |

**Notes**

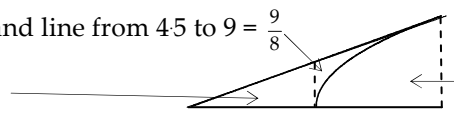
6. $\bullet^7$ may not be obvious until the final line of working and may be implied by final answer or a diagram.
7. At $\bullet^{11}$ the value of $x_A$ must lie between 0 and 9 exclusively, however, $\bullet^{12}$ and $\bullet^{13}$ are only available if $4\cdot5\le x_A<9$ .
8. Full marks are available to candidates who integrate with respect to $y$.

---

You may find the following helpful in marking this question:

Area between curve and line from 4·5 to 9 = $\frac{9}{8}$

Area of $\Delta_{SMALLER}=\frac{27}{8}$



Area of $\Delta_{LARGER}=13\cdot5$ or $\frac{27}{2}$

Area under curve from 4·5 to 9 = 9



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**7** (*a*) Given that $\log_4 x = \mathrm{P}$, show that $\log_{16} x = \frac{1}{2}\mathrm{P}$. **3**

**Generic Scheme** **Illustrative Scheme**

**(a)**

| | | | | |
|---|---|---|---|---|
| $\bullet^1$ | ss | convert from log to exponential form | $\bullet^1$ | $x = 4^{\mathrm{P}}$ |
| $\bullet^2$ | ss | know to and convert back to log form | $\bullet^2$ | $\log_{16} x = \log_{16} 4^{\mathrm{P}}$ |
| $\bullet^3$ | pd | process and complete | $\bullet^3$ | $\log_{16} x = \mathrm{P} \times \log_{16} 4$ and complete |

**Notes**

1. No marks are available to candidates who simply substitute in values and verify the result.

e.g. $\log_4 4 = 1$ and $\log_{16} 4 = \frac{1}{2}$ ✗

$\log_4 x = \mathrm{P}$ and $\log_{16} x = \frac{1}{2}\mathrm{P}$

**Regularly occurring responses**

**Response 1**

$\log_4 x = \mathrm{P}$

$x = 4^{\mathrm{P}}$ ✓ $\bullet^1$

$\log_{16} x = \frac{1}{2}P$ ✗ $\bullet^2$

$x = 16^{\frac{1}{2}\mathrm{P}}$

$= 4^{\mathrm{P}}$ ∧ $\bullet^3$

1 mark out of 3

**Response 2**

$\log_4 x = P$

$x = 4^P$ ✓ $\bullet^1$

$x^2 = 4^{2P}$

$= 16^P$ ✗ $\bullet^2$

$\log_{16} x^2 = P$

$2\log_{16} x = P$

$\log_{16} x = \frac{1}{2}P$ ✓ $\bullet^3$

2 marks out of 3

**Response 3**

$x = 4^{\mathrm{P}}$ ✓ $\bullet^1$

$x = \left(16^{\frac{1}{2}}\right)^{\mathrm{P}}$ or $16^{\frac{1}{2}\times \mathrm{P}}$ ✓ $\bullet^2$

$x = 16^{\frac{1}{2}\mathrm{P}}$

$\log_{16} x = \frac{1}{2}\mathrm{P}$ ✓ $\bullet^3$

Without this step $\bullet^2$ would be lost but $\bullet^3$ is still available as follow through.

3 marks out of 3

**Response 4**

$x = 4^{\mathrm{P}}$ ✓ $\bullet^1$

$\log_4 x = \mathrm{P}$

$4^{\log_4 x} = 4^{\mathrm{P}} = x$

$4^{\log_4 x} = x$

$4^{\mathrm{P}} = 16^{k\mathrm{P}}$

$4 = 16^k$ ✗ $\bullet^2$

$k = \frac{1}{2}$

$\therefore \log_{16} x = \frac{1}{2}\mathrm{P}$ as $16^{\frac{1}{2}} = 4$ ✓ $\bullet^3$

$\log_{16} x = k\mathrm{P}$

$16^{\log_{16} x} = 16^{k\mathrm{P}}$

$x = 16^{k\mathrm{P}}$

2 marks out of 3

**Response 5**

**Beware** that some candidates give a circular argument.

This is only worth $\bullet^1$.

$\log_4 x = \mathrm{P}$ then $\log_{16} x = \frac{1}{2}\mathrm{P}$

$x = 4^{\mathrm{P}}$ ✓ $\bullet^1$ $x = 16^{\frac{1}{2}\mathrm{P}}$

$\log_4 x = \log_4 4^{\mathrm{P}}$ $\log_{16} x = \log_{16} 16^{\frac{1}{2}\mathrm{P}}$

$\log_4 x = \mathrm{P}\log_4 4$ $\log_{16} x = \frac{1}{2}\mathrm{P}\log_{16} 16$

$\log_4 x = \mathrm{P}$ $\log_{16} x = \frac{1}{2}\mathrm{P}$ ✗

1 mark out of 3

**Response 6**

$\log_{16} x = \frac{\log_4 x}{\log_4 16} = \frac{\log_4 x}{2} = \frac{1}{2}\mathrm{P}$

✓ $\bullet^1$ ✓ $\bullet^2$ ✓ $\bullet^3$

Using change of base result.

3 mark out of 3



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| 7 (*b*) Solve $\log_3 x + \log_9 x = 12$. | 3 |
|---|---|

**Generic Scheme**

**(b)**

- $\bullet^4$ ss use appropriate strategy
- $\bullet^5$ pd start solving process
- $\bullet^6$ pd complete process via log to expo form

**Illustrative Scheme**

$\bullet^4$ $\log_3 x + \frac{1}{2}\log_3 x = 12$
$\bullet^5$ $\log_3 x = 8$
$\bullet^6$ $x = 3^8 \quad (= 6561)$

**or**

$\bullet^4$ $Q + \frac{1}{2}Q = 12$
$Q = 8$
$\bullet^5$ $\log_3 x = 8$
$\bullet^6$ $x = 3^8 \quad (= 6561)$

$\bullet^4$ $2\log_9 x + \log_9 x = 12$
$\bullet^5$ $\log_9 x = 4$
$\bullet^6$ $x = 9^4 \quad (= 6561)$

**or**

$\bullet^4$ $2Q + Q = 12$
$Q = 4$
$\bullet^5$ $\log_9 x = 4$
$\bullet^6$ $x = 9^4 \quad (= 6561)$

**Notes**

2. At $\bullet^4$ any letter except $x$ may be used in lieu of Q.
3. Candidates who use a trial and improvement technique by substituting values for $x$ gain no marks.
4. The answer with no working gains no marks.

**Regularly occurring responses**

**Response 1**

$Q + 2Q = 12$ ✓ $\bullet^4$ or ✗ $\bullet^4$
$3Q = 12$
$Q = 4$
$\log_3 x = 4$ ✗ $\bullet^5$ ✓ $\bullet^5$
$x = 3^4$ ✓ $\bullet^6$
$= 81$

2 marks out of 3

The marks allocated are dependent on what substitution is used for Q.

**Response 2**

$\log_3 x + 2\log_3 x = 12$ ✗ $\bullet^4$
$3\log_3 x = 12$
$\log_3 x = 4$ ✓ $\bullet^5$
$x = 3^4$ ✓ $\bullet^6$
$= 81$

2 marks out of 3

**Response 3**

$2\log_9 x + \log_9 x = 12$ ✓ $\bullet^4$
$\log_9 x^2 + \log_9 x = 12$
$\log_9 x^3 = 12$ ✓ $\bullet^5$
$x^3 = 9^{12}$
$x = \sqrt[3]{9^{12}}$
$x = 9^4$ ✓ $\bullet^6$
$x = 3^8$
$= 6561$

3 marks out of 3

**Response 4**

$\log_3 x = 8$ ✓✗ $\bullet^4$ ✓✗ $\bullet^5$
$x = 3^8$ ✓ $\bullet^6$
$= 6561$

Without justification, $\bullet^4$ and $\bullet^5$ are not available.



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